

Speech Tech Week 2

AQ2.6: Activity Questions 6 - Not Graded

1) Which of the following probability distributions has the shape of a bell curve?

1 point

- ☐ Binomial distribution
- ☐ Poisson distribution
- ☒ Gaussian distribution
- ☐ Bernoulli distribution

2) The presence of $(x - \mu)^2$ in the p.d.f of Gaussian distribution makes it _____.

1 point

- ☒ symmetric
- ☐ asymmetric

3) The Gaussian curve is always symmetrical about _____.

1 point

- ☐ 0
- ☐ 1
- ☒ mean
- ☐ standard deviation

mean

4) The value of derivative of a function at a maxima is _____.

0

✓

1 point

1) In the terminal analogous model for speech production, different mouth configurations correspond to different _____.

- ☐ lifters
- ☒ filters

filters

1 point

Yes, the answer is correct.

Score: 1

Accepted Answers:

filters

2) In the terminal analogous model for speech production, the random noise input is considered to represent _____ sounds.

1 point

- ☐ voiced
- ☒ unvoiced

unvoiced

Yes, the answer is correct.

Score: 1

Accepted Answers:

unvoiced

3) In STFT for speech applications, the usual overlap between adjacent windows is _____.

50

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

10 ms
window size = 25 ms

4) Psychophysical equivalent of frequency is _____

1 point

- ☐ intensity
☐ loudness
☒ pitch

intensity $\rightarrow$ loudness
frequency $\rightarrow$ pitch

Yes, the answer is correct.

Score: 1

Accepted Answers:

pitch

5) Apex part of the basilar membrane picks _____ frequencies.

1 point

- ☒ low
☐ high

Yes, the answer is correct.

Score: 1

Accepted Answers:

low



6) What is the equivalent of 1000 Hz frequency in the mel scale?

1000

Yes, the answer is correct.

Score: 2

Accepted Answers:

7) State True/False: Human ear is capable of performing time masking of the small intensity events that occurred just before a high intensity event.

1 point

- ☒ True
☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

eg. of pin dropped when compared to passing of an aeroplane

8) In cepstral domain, the vocal tract response is separated from the excitation by _____ filtering.

1 point

- ☒ low pass
☐ high pass
☐ band pass
☐ band stop

low pass

Yes, the answer is correct.

Score: 1

Accepted Answers:

low pass

9) In a probability density function (p.d.f), area under the curve must be equal to _____

1

Yes, the answer is correct.

Score: 1